

Trainings ▾

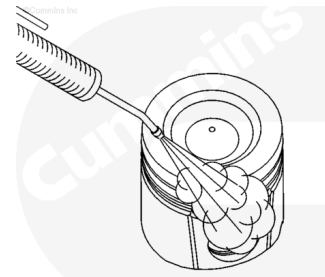
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ CAUTION ⚠

Do not use the bead blast method to clean the ring grooves or the skirt. The piston will be damaged by blast material embedded into the aluminum.



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Premium pistons have an anodized surface that is gray in appearance. Do **not** attempt to remove this surface when cleaning.

Make sure the piston rings have been removed.

Clean the outer layer of carbon from the piston with steam.

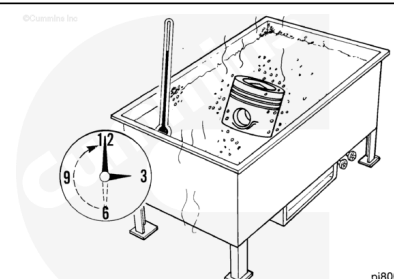
The bead blast method can be used to clean the top of the piston. If the bead blast method is used, cover the ring grooves and skirt securely.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

To reduce the possibility of engine damage, do not contaminate the wash tank solution with bead blast material.



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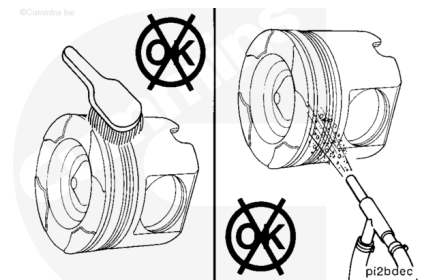
Allow the pistons to soak in the solvent for a minimum of 30 minutes.

For best results, soak the pistons for several hours, or overnight. Use a solvent that can be heated from 90°C to 95°C [180°F to 200°F]. Use a cleaning tank that will constantly mix and filter the solvent.

Use a kerosene emulsion-based solvent. Do **not** use a solvent that is higher than 9.5 pH. Do **not** use a solvent that contains chlorinated hydrocarbons with cresols, phenols, or cresylic components.

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⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not use a metal bristle brush. Damage to the piston ring grooves will result.

Clean the piston ring grooves with a brush or the gap end of a used piston ring.

Repeat the soaking and scrubbing process until the piston is cleaned thoroughly.

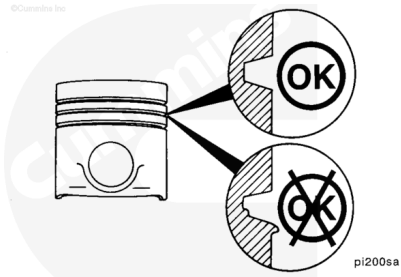
Rinse the solvent from the piston with steam.

Dry the piston with compressed air.

Inspect the ring grooves.

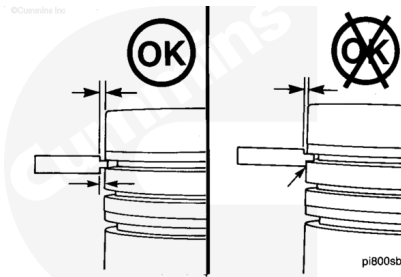
If there is a visible ridge in the back of the ring groove, the piston **must** be replaced.

If a lip has formed on the outside diameter of the groove, the piston **must** be replaced.



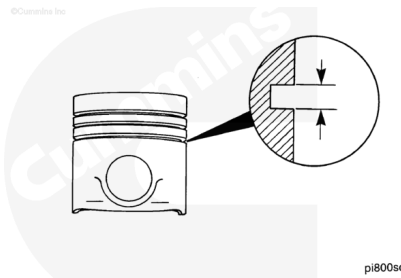
Measure both of the piston compression ring grooves with the piston ring gauge, Part Number ST-560

If the measurements are **not** within specifications, the piston **must** be replaced.



Measure the oil ring groove.

Piston Oil Ring Groove Width		
mm		in
4.724	MIN	0.186
4.826	MAX	0.190

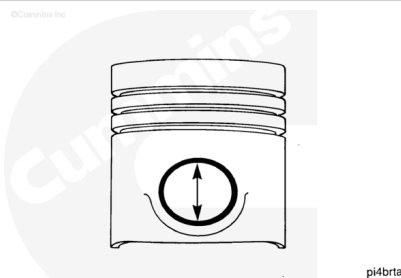


If the piston is **not** within specifications, it **must** be replaced.

Measure the piston pin bore when the room temperature is approximately 21°C [70°F].

Measure and record the inside diameter of the piston pin bore and the points illustrated in the graphic.

Piston Pin Bore Inside Diameter		
mm		in
60.942	MIN	2.3993
60.952	MAX	2.3997



If the piston pin bore is **not** within specifications, the piston **must** be replaced.

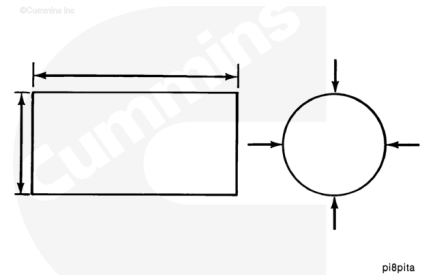
Measure the piston pin outside diameter and length.

Piston Pin Outside Diameter

mm		in
60.929	MIN	2.3988
60.935	MAX	2.3990

Piston Pin Length

mm		in
132.588	MIN	5.220
132.842	MAX	5.230

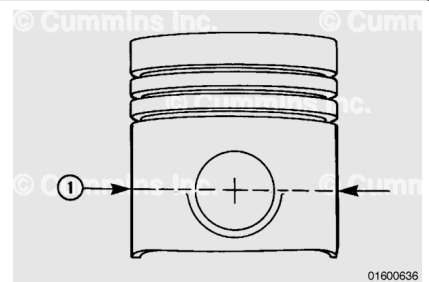


Check each piston for scratches or indications of scuffing. Reference Part Reuse Guidelines, Bulletin 3810303, for details.

The piston measurement is to be recorded at room temperature of approximately 21°C [70°F].

Piston Skirt Outside Diameter

	mm		in
Gallery Cooled	158.521	MIN	6.241
	158.549	MAX	6.242
Under Crown Cooled	158.549	MIN	6.242
	158.572	MAX	6.243



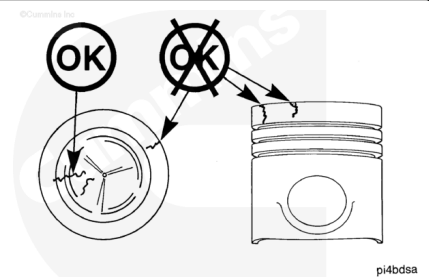
Note : Gallery cooled pistons are to be measured 25.4 mm [1.00 in] from the bottom of the piston skirt.

Note : Under crown cooled pistons are to be measured 51 mm [2.00 in] from the bottom of the piston skirt.

Crack Detection Kit, Part Number 3375432, contains cleaner, penetrant, and developer necessary to check for cracks using the dye penetrant method.

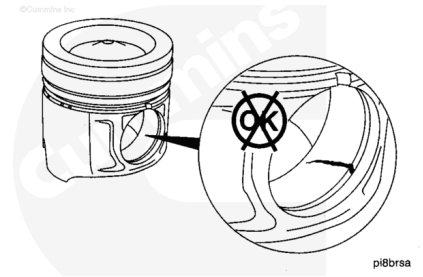
Check for cracks on the top of the piston and in the piston pin bore with Crack Detection Kit, Part Number 3375432, or equivalent.

The piston **must** be replaced if there is a crack on the top of the piston that extends more than half the distance across the shaded area, or a crack on the top of the piston that extends



over the rim and toward the ring grooves.

The piston **must** be replaced if there is a crack in the piston pin bore.



Inspect the piston for scratches or indications of scuffing.
Reference the Parts Reuse Guidelines, Bulletin 3810303.

